

How much energy does it take?

Work, thermal energy and specific heat capacity

Physics · 3A · 9 October 2026

Where we start from

(slide 2)

On 6 October we said one thing, but a fundamental one: the temperature of a substance is related to the **average kinetic energy** of its particles. Increasing the average kinetic energy means that the particles move faster, or vibrate more.

And what does an oven do to cook food? It transfers thermal energy: it agitates the particles. The oven agitates them with a flame. But is there another way to agitate them, one that has nothing to do with a flame?

Agitating particles without a flame

(slide 3–4)

Take a syringe full of air, with insulated walls: the gas cannot lose energy to the surroundings. I push the piston in very fast.

Before moving on. What happens to the temperature of the gas: does it go down, stay the same, or go up?

It goes up. Let us look at what the particles do (figure 1). A particle that hits the piston while it moves towards it bounces back **faster**, like a ball hit by a racket. The piston has done **work** on the gas: the average kinetic energy of the particles increases, and so does the temperature. No flame, no contact with anything hot.

If I pull the piston out, the opposite happens: the particles slow down and the gas cools. It is the same reason why a spray can gets cold as the gas expands, and why a bicycle pump gets warm under the piston.

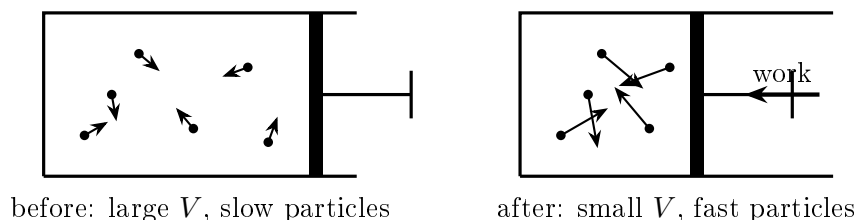


Figure 1: Pushing the piston on an insulated gas: arrows show particle velocities (schematic, not to scale). The model is two-dimensional with no collisions between particles: in a real gas the temperature rises much less.

Can you cook a chicken by slapping it?

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If agitating particles mechanically warms things up, then I should be able to cook something that way. For example a chicken, by slapping it. (No chickens were harmed in this experiment.)

A chicken at room temperature is at $T = 20^\circ\text{C}$; cooked through, it is at about 70°C .

Before moving on. *In principle*, can slaps raise the temperature of a chicken by 50 K? Yes or no? Write down your answer: we come back to it at the end of the chapter.

To answer we need two things: to understand what a slap does to the particles, and to work out how much energy is needed.

Inside the slap

(slide 6)

Before moving on. When the hand stops against the chicken, where does its kinetic energy go?

It does not disappear. Before the impact, all the particles of the hand move together in the same direction: **ordered** motion. Afterwards, the particles of the chicken move in all directions: **random** motion (figure 2). The macroscopic kinetic energy becomes **thermal energy**. This is an **inelastic collision**: total energy is conserved, macroscopic kinetic energy is not. Each slap adds a little, and T rises slap after slap.

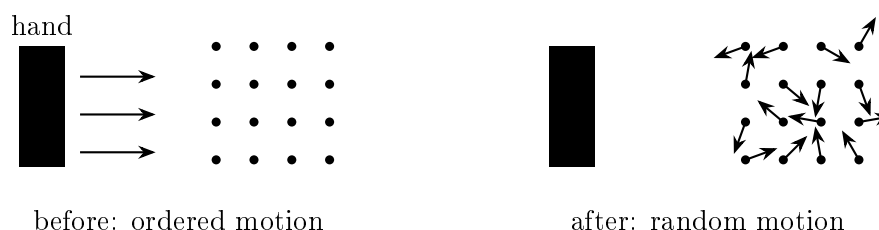


Figure 2: Arrows show particle velocities (schematic, not to scale).

Joule's paddle wheel

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We have seen that mechanical work can warm things up. Can we measure how much work is needed for a given temperature rise? James Joule did it around 1840 with an apparatus (figure 3): two falling weights, two pulleys, and a paddle turning in an insulated container full of water. The weights lose gravitational potential energy; the paddle stirs the water, that is, it does work on it.

For each fall through a height h of two weights of mass m each:

$$\Delta E_p = 2 m g h.$$

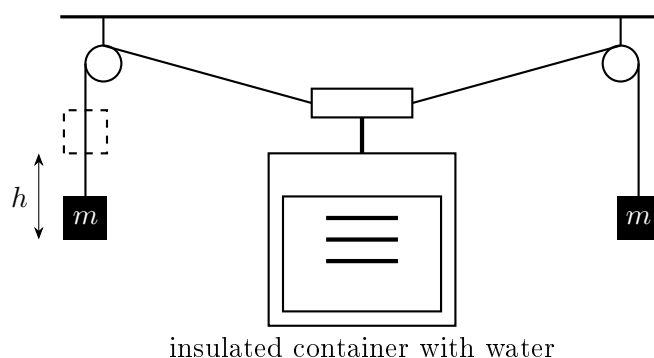


Figure 3: Joule's apparatus (schematic): the falling weights turn the paddles, which stir the water.

Before moving on. Ten falls with two 5 kg weights, $h = 1.0$ m, $g = 10$ N/kg, on 100 g of water. By how much does the temperature rise?

Solution. Work for one fall: $2 \cdot 5 \cdot 10 \cdot 1 = 100$ J. For ten falls: $W = 1\,000$ J. If all the work becomes thermal energy of the water and $c = 4\,186$ J/(kg K):

$$\Delta T = \frac{W}{mc} = \frac{1000}{0.100 \times 4186} \approx 2.4 \text{ K}.$$

If we change the mass of the weights, the ratio $W/\Delta T$ stays constant: it is 419 J/K for 100 g of water. More work, proportionally larger ΔT . That number has a name, which we meet shortly.

Joule's result: 4.186 J warm 1 g of water by 1 K, that is, one calorie. In the IGCSE syllabus only the joule is used. The simulation we use in class is ideal: no friction and no loss of energy (in a real experiment the temperature rises are tiny and very noisy).

Constant power: time measures energy

(slide 8)

A heater has a power P in watts, that is, joules per second. If it stays on for a time t , it transfers to the water an energy

$$Q = Pt,$$

where Q is the energy transferred (J), P the constant power (W) and t the time (s). A 500 W heater on for 60 s: $Q = 500 \times 60 = 30\,000$ J = 30 kJ.

On 6 October, on the heating curve, the horizontal axis was time. At constant power, time is just a way of measuring the energy supplied.

Before moving on. If I double the power, what changes in the graph of T against time? And in the graph of T against energy?

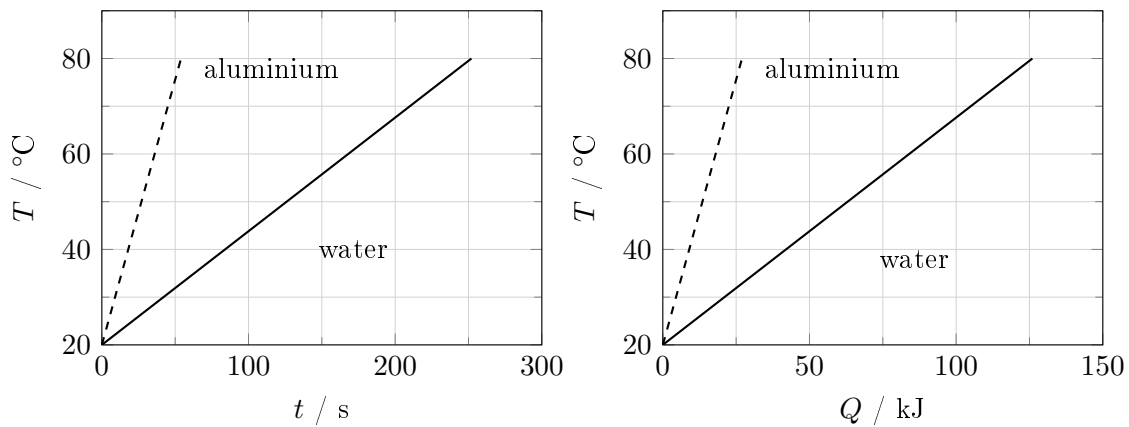


Figure 4: $T(t)$ on the left and $T(Q)$ on the right, $P = 500 \text{ W}$, $m = 0.5 \text{ kg}$, from 20°C to 80°C . No change of state, no losses.

From $T(t)$ to $T(Q)$

(slide 9)

Figure 4 compares the two graphs for 0.5 kg of water and 0.5 kg of aluminium, with $P = 500 \text{ W}$. At this power they are identical: only the label of the horizontal axis changes, because $Q = Pt$.

Before moving on. I lower the power. What moves in the graph on the left, and what in the one on the right? Then I double the mass: what happens to the slope of $T(Q)$? And if I change the substance?

With less power the left curve flattens: it takes more time. The right curve **does not move**: $T(Q)$ depends on the substance and on the mass, not on how fast I heat (in the ideal model, with no losses). If the mass doubles, the slope halves. For the same mass, the larger the c of the substance, the less steep the line. The slope of $T(Q)$ has a name, and so does its inverse.

Thermal capacity and specific heat capacity

(slide 10–11)

Note: thermal capacity and specific heat capacity are not in the IGCSE 0653 syllabus (they are in 0625). But they are the tool for doing the calculations.

The slope of $T(Q)$ is $\frac{\Delta T}{Q} = \frac{1}{C}$.

Thermal capacity: $C = \frac{Q}{\Delta T}$, in J/K . It is the energy needed to raise the temperature of the whole object by 1 K .

Specific heat capacity: $c = \frac{Q}{m \Delta T}$, in J/(kg K) . It is the energy needed to raise the temperature of 1 kg of a substance by 1 K . It does not depend on the object, but on the substance.

$$Q = m c \Delta T \qquad C = m c$$

Check: for 0.5 kg of water $C = 0.5 \times 4200 = 2100 \text{ J/K}$, that is 2.1 kJ/K: the slope of $T(Q)$ in the figure is $1/2.1 = 0.48 \text{ K/kJ}$. Since a change of 1 K equals a change of 1°C , the unit can also be written $\text{J}/(\text{kg } ^\circ\text{C})$. The heat Q is energy *transferred* thermally: it is not possessed.

Substance	$c / \text{J}/(\text{kg K})$
Water	4200
Chicken (estimate, mostly water)	3000
Olive oil	2000
Aluminium	900
Copper	390

Typical values at room temperature; the value for chicken is an estimate.

Same energy, different substances

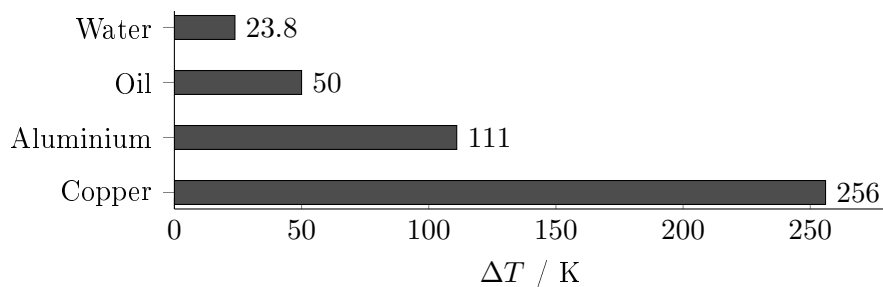
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Before moving on. I supply 100 kJ to 1 kg of each substance, all at 20°C . Which one ends up the hottest?

From the slope: the substance with the smallest C has the steepest curve and rises the most.

With $\Delta T = \frac{Q}{mc}$:

Substance	$c / \text{J}/(\text{kg K})$	$\Delta T / \text{K}$
Water	4200	$100\,000/4200 = 23.8$
Olive oil	2000	$100\,000/2000 = 50$
Aluminium	900	$100\,000/900 = 111$
Copper	390	$100\,000/390 = 256$



Copper would reach 276°C , still below its melting point, so the model holds. For the same energy and mass, ΔT is inversely proportional to c . A small c means heating up quickly (and cooling down quickly); a large c means **thermal inertia**.

Back to the chicken: how many slaps?

(slide 14)

One kilogram of chicken from 20 °C to 70 °C. One slap: a hand of about 0.5 kg reaching about 5 m/s.

1. $\Delta T = 70 - 20 = 50$ K. First the temperature change.
2. $Q = m c \Delta T = 1 \times 3000 \times 50 = 150\,000$ J = 150 kJ. This is the *minimum* energy: no losses at all.
3. $E_k = \frac{1}{2} m v^2 = \frac{1}{2} \times 0.5 \times 5^2 = 6.25$ J: kinetic energy of one slap, all of it dissipated.
4. $N = \frac{150\,000}{6.25} = 24\,000 \approx 2 \times 10^4$ slaps. *At least* this many.

(The values $c \approx 3000$ J/(kg K), $m_{\text{hand}} \approx 0.5$ kg and $v \approx 5$ m/s are reasonable estimates, not measurements: if they change a little, the order of magnitude, 10^4 , stays the same.)

Someone actually did it

(slide 15)

The YouTuber Louis Weisz built a chicken-slapping machine in 2021. The target was 55 °C–60 °C, held for at least an hour. As reported by ScienceAlert, it took **at least 135 000 slaps**, up to 8 hours, about 7.5 kW h.

Before moving on. Why is the real number so much larger than our minimum of about 24 000?

Because our calculation is a model with no losses, and it only *reaches* 70 °C. In the real machine energy is lost to sound, air and the machine itself, and then the temperature must be *maintained* for an hour. An order of magnitude: 7.5 kW h is 27 MJ, about 180 times the 150 kJ of our calculation.

Go back to the answer you wrote down at the start: in principle, and in practice too, a chicken can be cooked by slapping it.

Source: ScienceAlert, April 2021, on Louis Weisz's video. Figures as reported in the article.

Water and the planet

(slide 16–18)

We have a tool, $Q = m c \Delta T$. What has an enormous thermal capacity on our planet?

Before moving on. How thick must a layer of ocean be to have the same thermal capacity as the whole atmosphere: 350 m, 3.5 m or 3.5 km?

Solution. Thermal capacity of the atmosphere, $C = m c$:

$$C_{\text{atm}} = 5.15 \times 10^{18} \times 1005 \approx 5.2 \times 10^{21} \text{ J/K.}$$

Thermal capacity of an ocean layer of thickness h (area A , density ρ):

$$C_{\text{sea}} = A h \rho c \approx 1.5 \times 10^{21} h \text{ J/K} \quad (h \text{ in metres}).$$

Equating the two:

$$h = \frac{5.2 \times 10^{21}}{1.5 \times 10^{21}} \approx 3.5 \text{ m.}$$

A few metres of sea are worth the whole air above our heads.

Average data: mass of the atmosphere 5.15×10^{18} kg, c of air 1005 J/(kg K); ocean surface area 3.6×10^{14} m², density 1025 kg/m³, $c \approx 3990$ J/(kg K). These are orders of magnitude. Calculation by the teacher.

Where does the excess energy go?

(slide 19–20)

The greenhouse effect traps more energy than the planet emits. Where does it go? Into the ocean: about **91 %** of the excess energy (since the 1970s); everything else, air, land and ice, takes the remaining 9 %. (Source: IPCC AR6, Working Group I, FAQ 7.1.) With an enormous C , the sea absorbs a great deal of energy with a small change in T , because $\Delta T = Q/C$: this is why air temperature alone underestimates what is happening.

A warming ocean, even slightly, has cascading consequences:

- the water **expands**: sea level rises (thermal expansion);
- more **evaporation**: the water cycle intensifies, with droughts and extreme rain;
- warm surface water **fuels tropical cyclones**;
- warm and cold layers mix less: **less oxygen** and fewer nutrients at depth;
- the energy content of the oceans set a **record** every year from 2019 to 2022, also in the Mediterranean.

Source: Scientific American, on the report by L. Cheng and colleagues (2022 data).

A question from the beach

(slide 21)

Before moving on. Why is the sea still warm in October, when the air is already cool?

The sea has an enormous thermal capacity: during the summer it stored a great deal of energy with only a small change in T . To cool down it must *release* the same energy, and it does so slowly: $\Delta T = Q/C$. Same principle, planetary scale: the oceans **remember** warming much longer than the air does. (The teacher's own reasoning, not taken from a source: the cause, the excess energy, keeps acting, so waiting for the sea to cool is not enough.)

And then, the energy moves

(slide 22)

A warm sea and a cold air: how does energy pass from one to the other? In three ways:

- **conduction**: in solids, vibration after vibration;
- **convection**: in fluids, through differences in density (breezes, currents, high and low pressure);

- **radiation**: no matter needed, as from the Sun.

Monday 12 October: the three modes, one by one.

Three points to keep

(slide 23)

1. Mechanical **work** (slap, piston, paddle wheel) and thermal transfer raise the average kinetic energy of the particles in the same way: T rises.
2. Warming needs energy: $Q = m c \Delta T$. The slope of $T(Q)$ is $1/C$, with $C = m c$.
3. Water has an enormous mass and c : the oceans absorb 91 % of the excess energy and release it slowly.

Exercises

(slide 24)

1. How much energy is needed to heat 0.50 kg of water from 20 °C to 80 °C?
2. How much energy is needed to heat 200 g of aluminium from 20 °C to 70 °C?
3. A 500 W heater warms 1.0 kg of water. How long does it take for the temperature to rise by 10 K?
4. Explain, using the words *thermal capacity*, *energy* and *cooling*, why the sea is still warm in October.

Solutions

1. $\Delta T = 80 - 20 = 60 \text{ K}$; $Q = m c \Delta T = 0.50 \times 4200 \times 60 = 126\,000 \text{ J} = 126 \text{ kJ}$.
2. $m = 0.200 \text{ kg}$; $\Delta T = 70 - 20 = 50 \text{ K}$; $Q = 0.200 \times 900 \times 50 = 9\,000 \text{ J} = 9.0 \text{ kJ}$.
3. $Q = 1.0 \times 4200 \times 10 = 42\,000 \text{ J}$; $t = Q/P = 42\,000/500 = 84 \text{ s}$.
4. The sea has a large thermal capacity. During the summer it stored energy with a small change in temperature. To cool down it must release the same energy, so it does so slowly: $\Delta T = Q/C$.